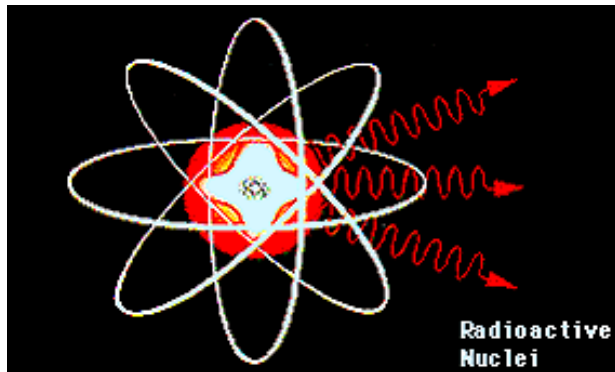
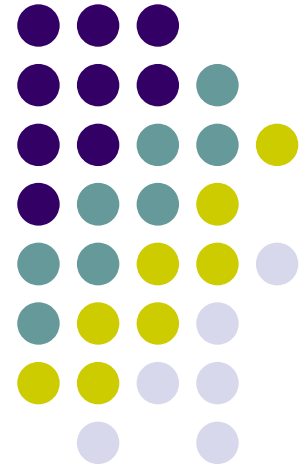
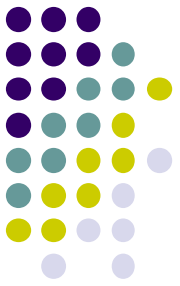




Radiopharmaceutics



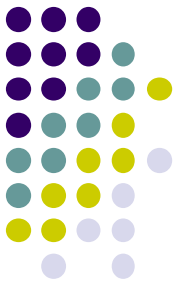
What is Radiopharmacy?



Radiopharmacy = Nuclear Pharmacy

Nuclear pharmacy is a specialty area of pharmacy practice dedicated to the compounding and dispensing of radioactive materials for use in nuclear medicine procedures.”

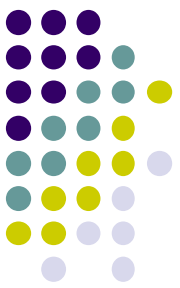




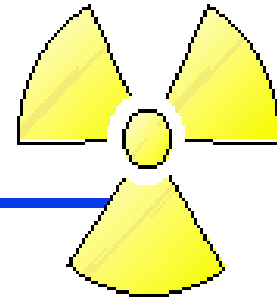
Introduction:

- All substances are made of **atoms**.
- These have **electrons** (e) around the outside (negatively charged),

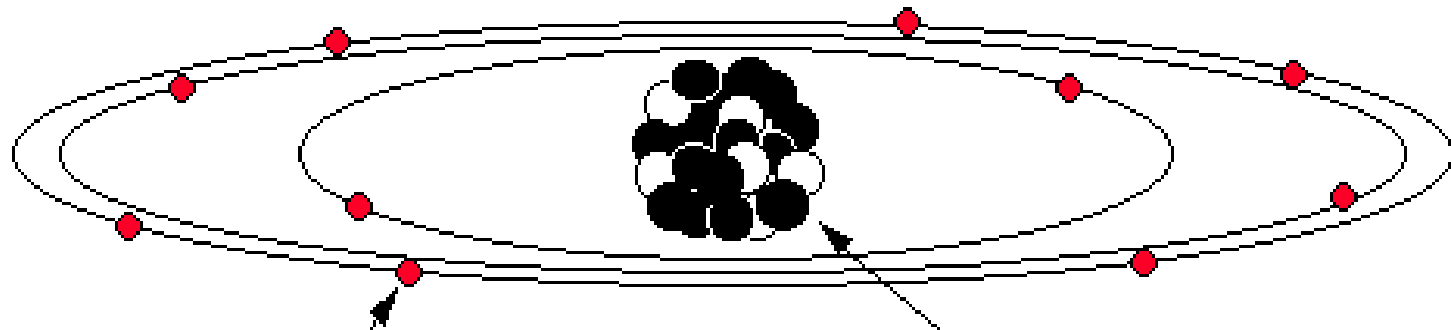
and a **nucleus** in the middle.
- The nucleus consists of **protons** (positively charged) and **neutrons** (neutral).
- **The atomic number** of an atom is the number of protons in its nucleus.
- **The atomic mass** is the number of protons + neutrons in its nucleus.



The Atom



Example - Neon-20



Electrons

The Nucleus which
contains neutrons and
protons



Introduction:

- *Isotopes* of an atom have the same number of protons, but a different number of neutrons.

- *Example:*

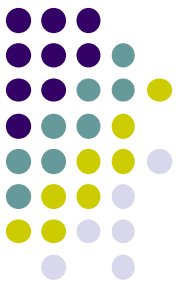
Consider a carbon atom:

It has 6 protons and 6 neutrons - we call it "carbon-12" because it has an atomic mass of 12 (6 plus 6).

One useful isotope of carbon is "carbon-14", which has 6 protons and 8 neutrons.

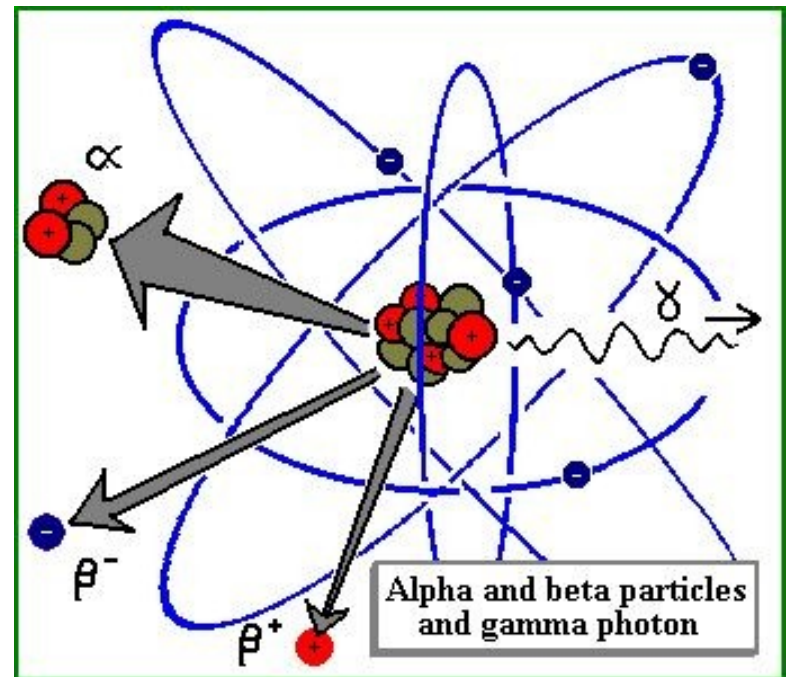
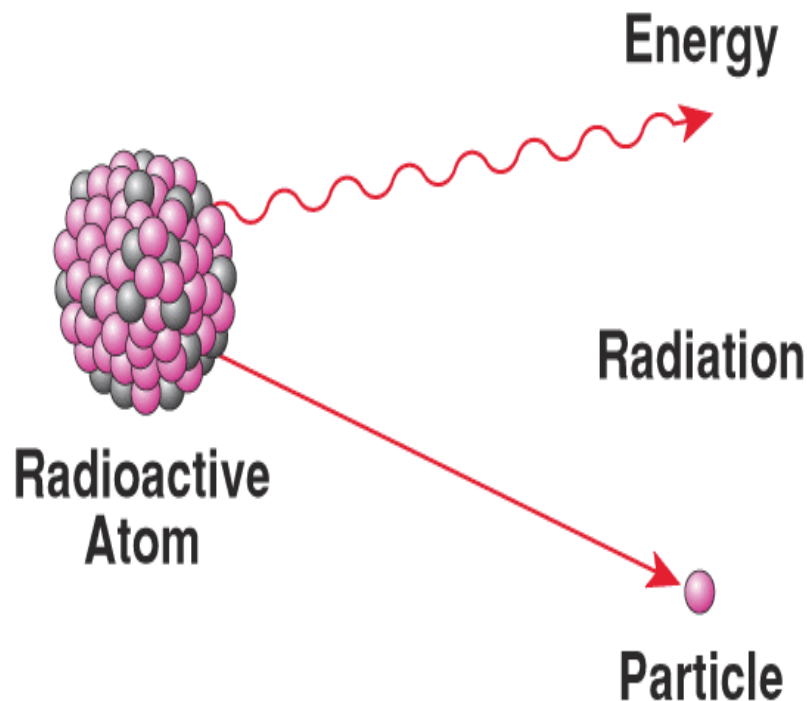
Radioisotopes, Radionuclides: unstable isotopes which are distinguishable by radioactive transformation.

Radioactivity: the process in which an unstable isotope undergoes changes until a stable state is reached and in the transformation emits energy in the form of radiation (alpha particles, beta particles and gamma rays).



Introduction:

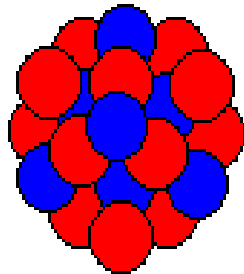
- *Radiation* refers to particles or waves coming from the nucleus of the atom (**radioisotope or radionuclide**) through which the atom attempts to attain a more stable configuration.





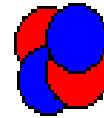
Radioactivity

heavy, unstable
element (e.g. Uranium)



spontaneous
decay

alpha particles (He nuclei)



gamma ray

proton



beta particle (electron)



neutron



Types of radioactivity:

How to produce a radioactive nuclide ?



1- Natural radioactivity:

Nuclear reactions occur spontaneously

2- Artificial radioactivity:

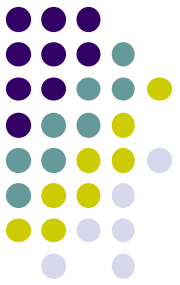
The property of radioactivity produced by particle bombardment or electromagnetic irradiation.

A- Charged-particle reactions

e.g. protons (${}^1_1\text{H}$)

e.g. deuterons (${}^2_1\text{H}$)

e.g. alpha particles (${}^4\text{He}$)



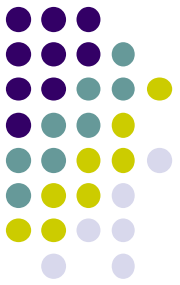
Types of radioactivity:

B- Photon-induced reactions

The source of electromagnetic energy may be gamma-emitting radionuclide or high-voltage x-ray generator.

C- Neutron-induced reactions

- It is the most widely used method
- It is the bombardment of a nonradioactive target nucleus with a source of thermal neutrons.



Production of radionuclides:

1- Charged particle bombardment

Radionuclides may be produced by bombarding target materials with charged particles in **particle accelerators such as cyclotrons.**

- **A cyclotron consists of :**

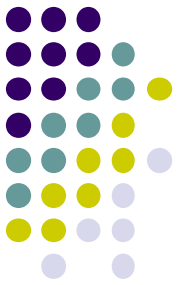
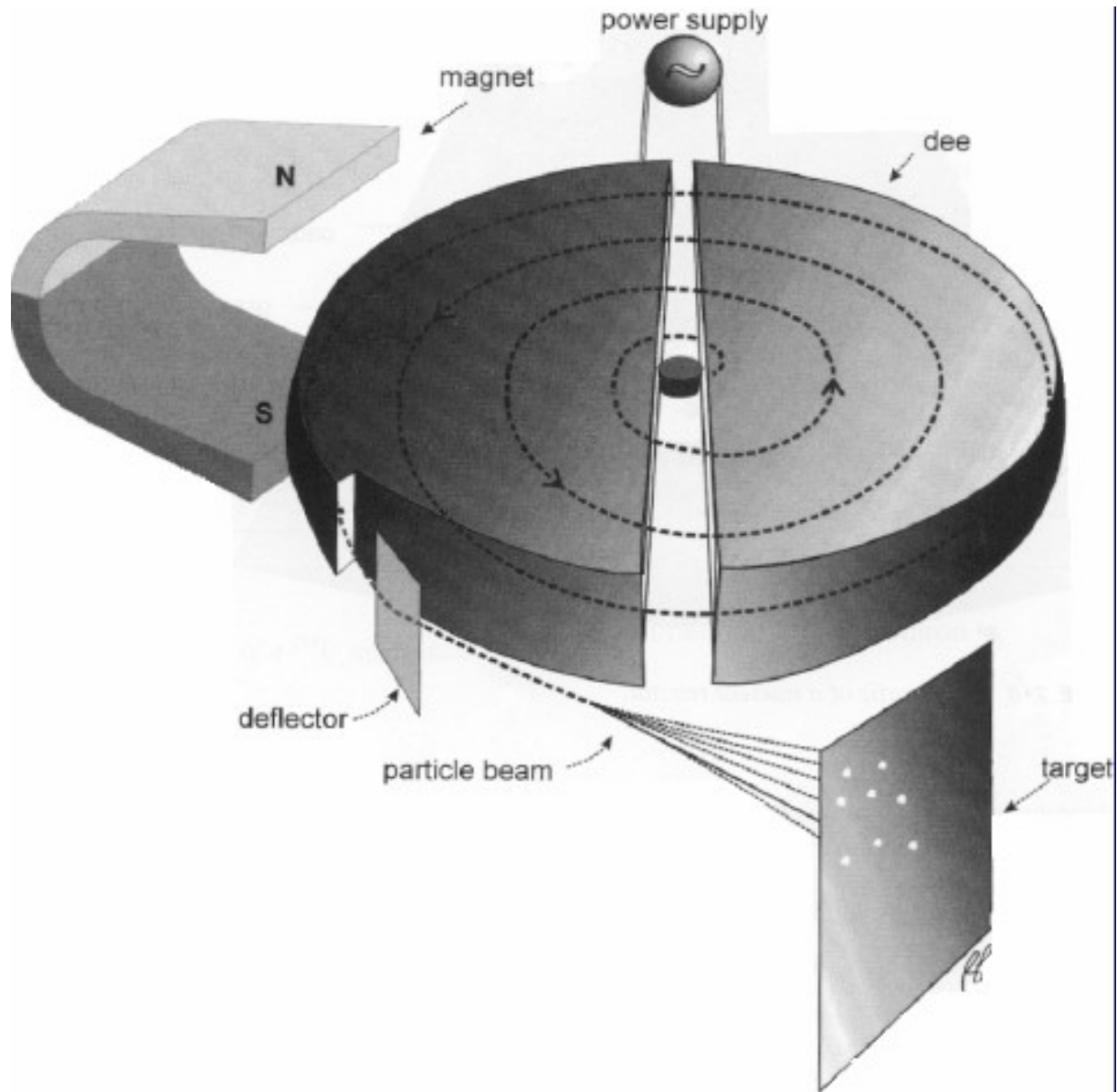
Two flat hollow objects called dees.

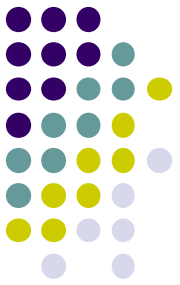
The dees are part of an electrical circuit.

On the other side of the dees are large magnets that (drive) steer the injected charged particles (**protons, deuterons, alpha and helium**) in a circular path

The charged particle follows a circular path until the particle has sufficient energy that it passes out of the field and interact with the target nucleus.

Cyclotron



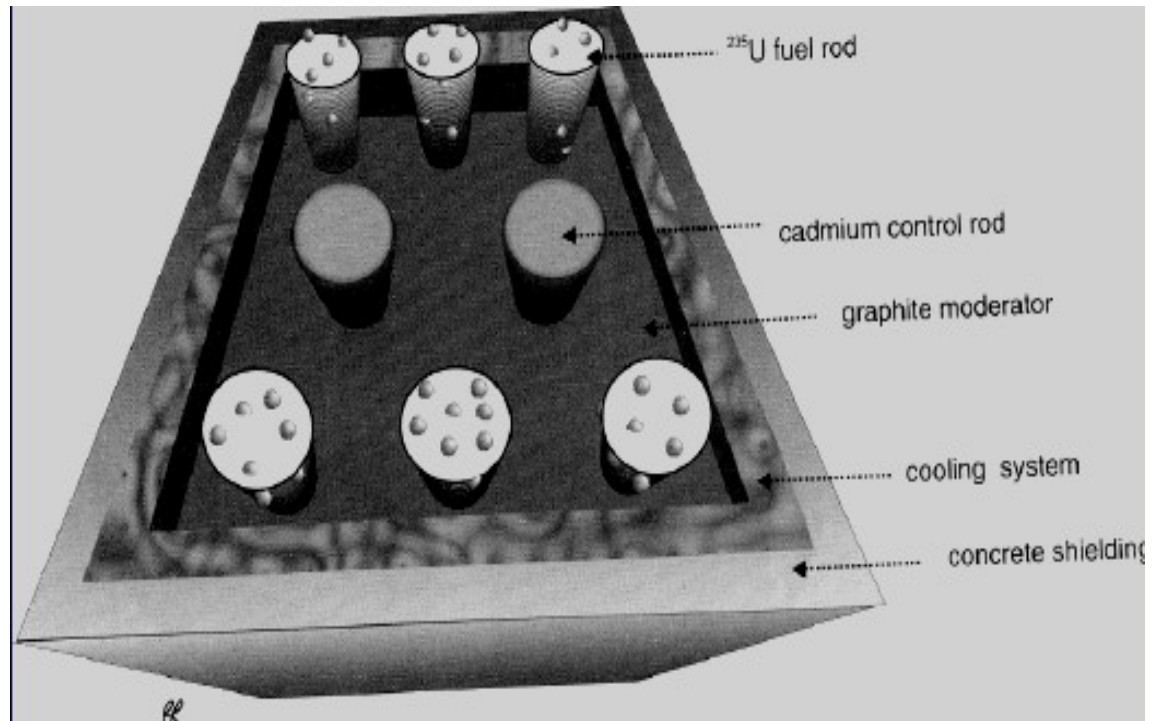


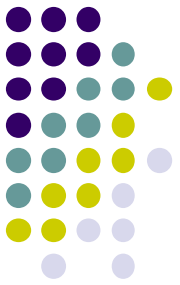
Production of radionuclides:

2- Neutron bombardment

Radionuclides may be produced by bombarding target materials with neutrons in **nuclear reactors**

- The majority of radiopharmaceuticals are produced by this process





Production of radionuclides: :

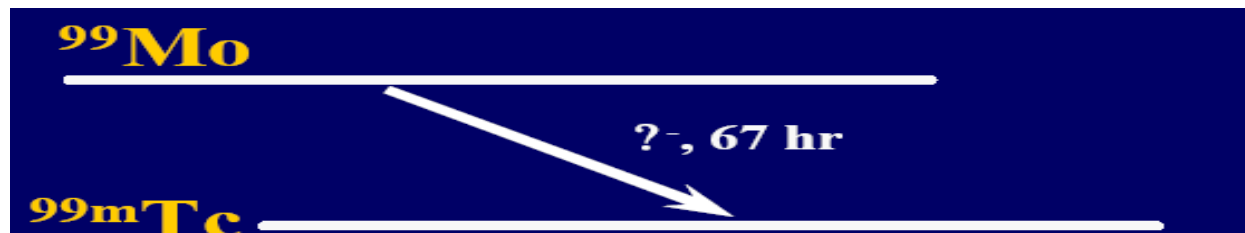
3- Radionuclide generator systems

- *Principle:*

A long-lived parent radionuclide is allowed to decay to its short-lived daughter radionuclide and the latter is chemically separated in a physiological solution.

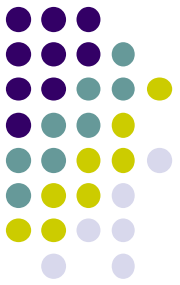
Example:

- technetium-99m, obtained from a generator constructed of molybdenum-99 absorbed to an alumina column.



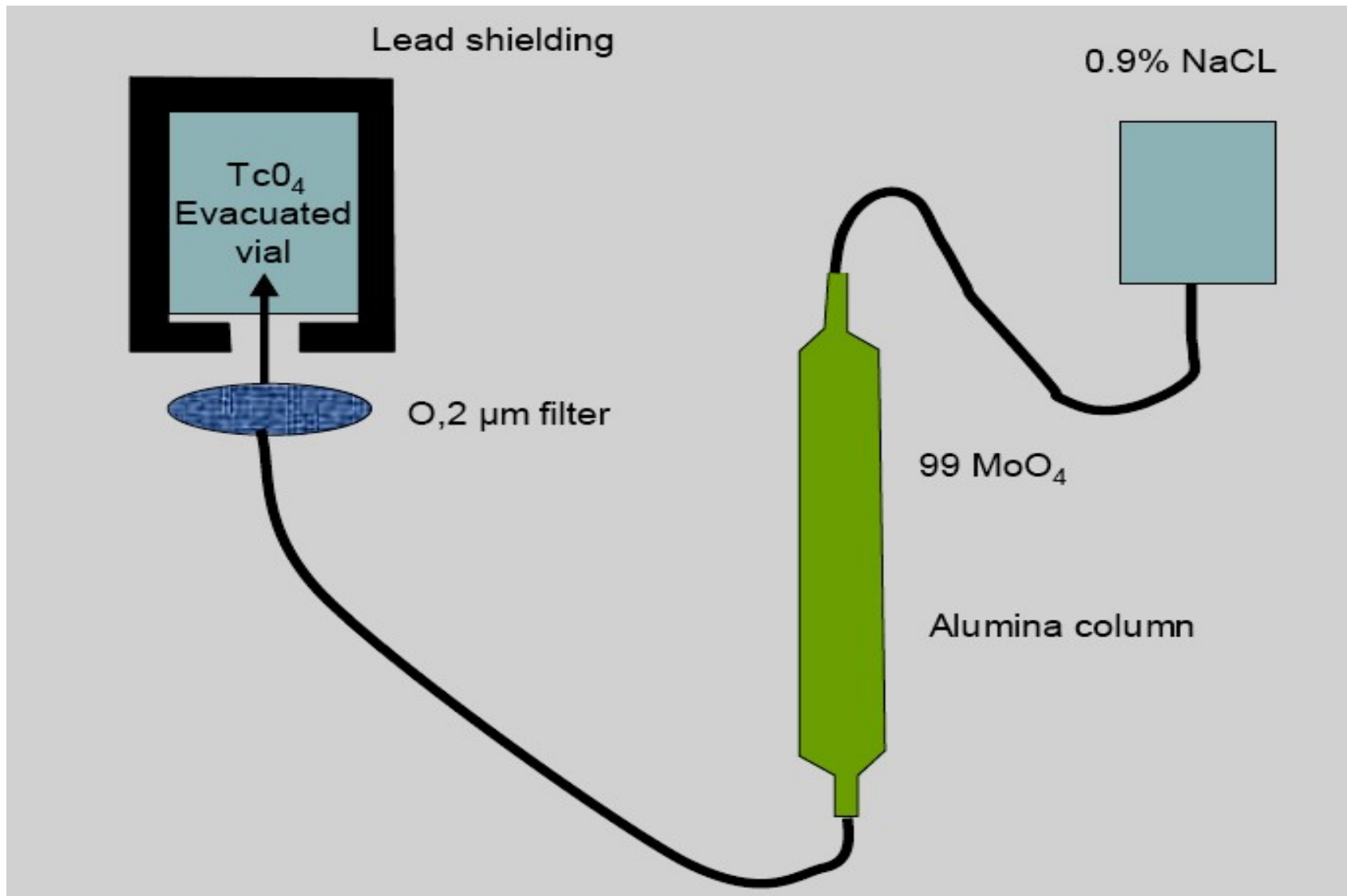
Eluted from the column with normal saline

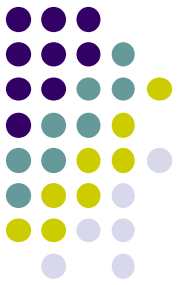
99Mo/99mTc Generator:



- *Parent*: 99Mo as molybdate ($^{99}\text{MoO}_4^{-2}$)
- *Half-life*: 66 hr.
- *Decays by* β^- - emission, gamma: 740, 780 keV.
- High affinity to alumina compared to 99mTc.
- *Daughter*: $^{99\text{m}}\text{Tc}$ as pertechnetate ($^{99\text{m}}\text{TcO}_4^{-1}$)
- *Adsorbent Material*: Alumina (aluminum oxide, Al_2O_3)
- *Eluent*: saline (0.9% NaCl)
- *Eluate*: ($^{99\text{m}}\text{TcO}_4^{-1}$)

$^{99}\text{Mo}/^{99\text{m}}\text{Tc}$ Generator





Radioactive decay:

- The rate of decay can be described by:*

$$N = N_0 e^{-\lambda t}$$

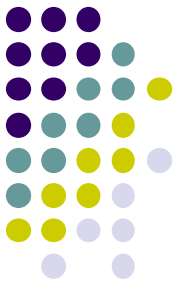
where N is the number of atoms at elapsed time t , N_0 is the number of atoms when $t = 0$, and λ is the disintegration constant characteristic of each individual radionuclide.

$$T_{1/2} = 0.693 / \lambda$$

The intensity of radiation can be described by

$$I = I_0 e^{-0.693 / T_{1/2} t}$$

Radioactive Decay Law

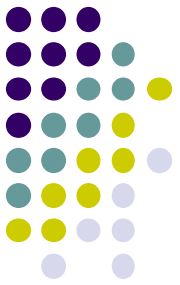


*The rate at which a particular **decay** process occurs is proportional to the (N) number of radioactive nuclei present*

decay rate

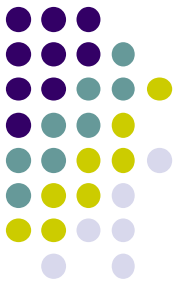
$$\frac{dN}{dt} = -\lambda N$$

$$N = N_0 e^{-\lambda t}$$



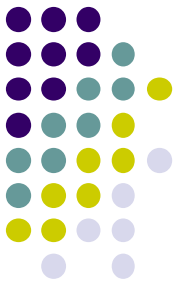
Radioactive decay:

- **Half life** — symbol $t_{1/2}$ — the time taken for the activity of a given amount of a radioactive substance to decay to half of its initial value.
- **Total activity** — symbol A — number of decays an object undergoes per second.
- **Radionuclidic purity-** is that percentage of the total radioactivity that is present in the form of the stated radionuclide.



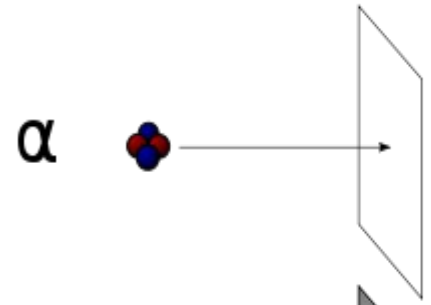
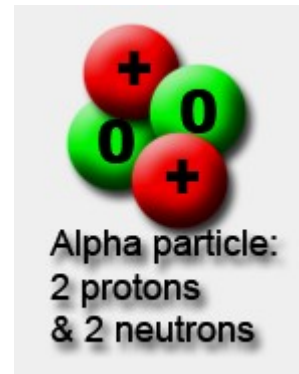
Mode of radioactive decay:

- ***Radioactive decay*** is the process in which an unstable atomic nucleus spontaneously loses energy by emitting ionizing particles and radiation.
- This decay, or loss of energy, results in an atom of one type, called the ***parent nuclide*** transforming to an atom of a different type, named the ***daughter nuclide***.
- ***When an unstable nucleus decays, It may give out:-***

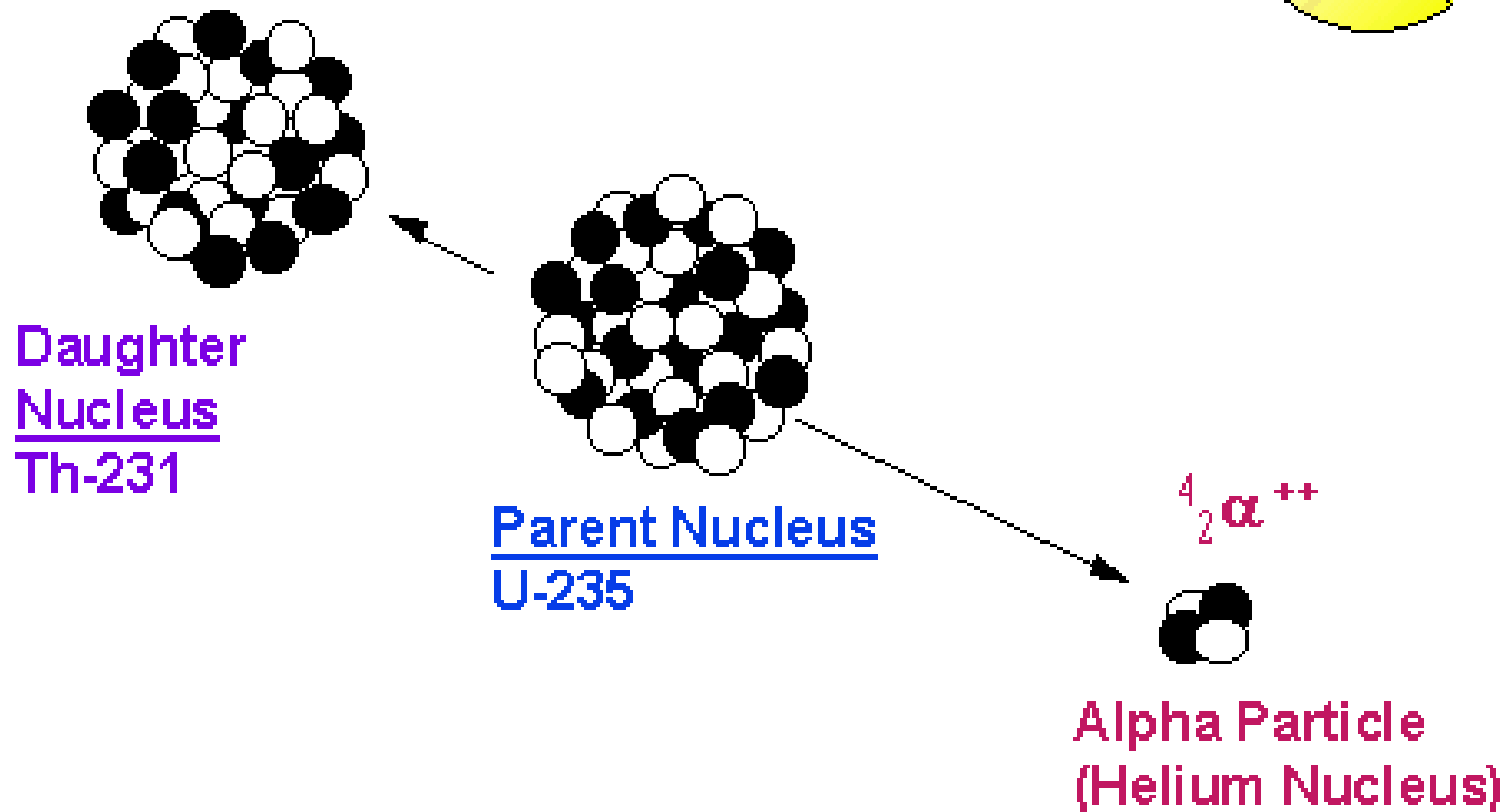
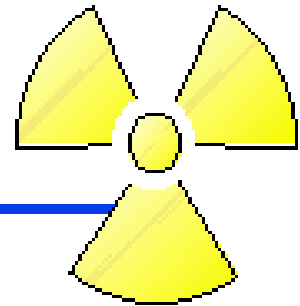


1- Alpha particle decay:

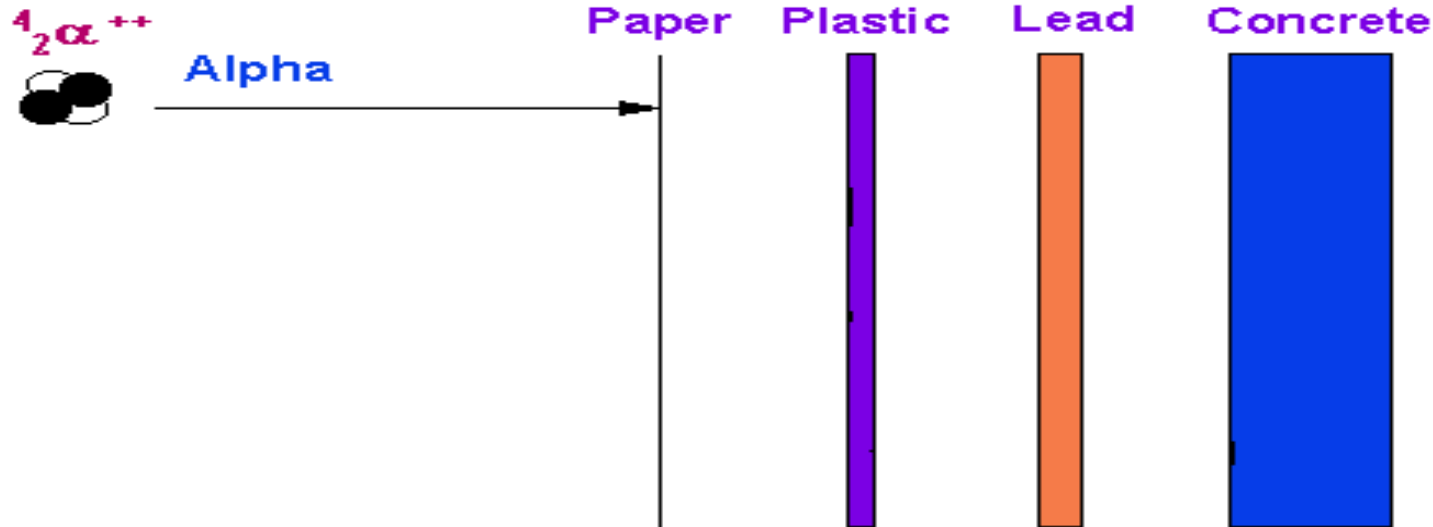
- Alpha particles are made of **2 protons and 2 neutrons**.
- We can write them as ${}^4_2\alpha$, or ${}^4_2\text{He}$, because they're the same as a helium nucleus.
- This means that when a nucleus emits an alpha particle, its atomic number decreases by 2 and its **atomic mass decreases by 4**.
- Alpha particles are relatively **slow** and **heavy**.
- They have a **low penetrating power** - you can stop them with just a sheet of **paper**.
- Because they have a large charge, alpha particles ionise other atoms strongly.
- Alpha-decay occurs in very heavy elements, for example, **Uranium and Radium**.



Alpha Particle Radiation



Penetrating Distances

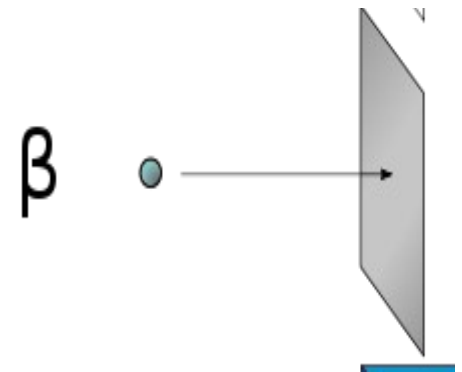
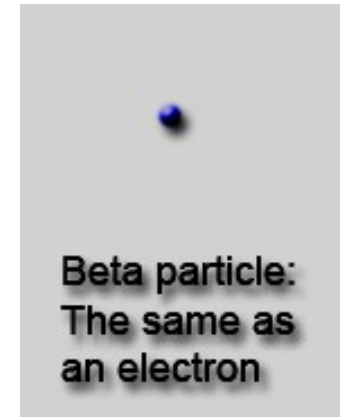
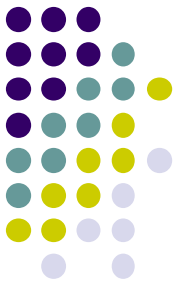


Since alpha particles cannot penetrate the dead layer of the skin, they do not present a hazard from exposure external to the body.

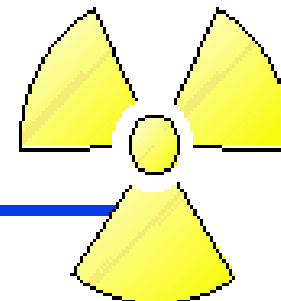
However, due to the **very large number of ionizations** they produce in a very short distance, alpha emitters can present a serious hazard when they are in close proximity to cells and tissues such as the lung. Special precautions are taken to ensure that **alpha emitters are not inhaled, ingested or injected.**

2- Beta particle decay:

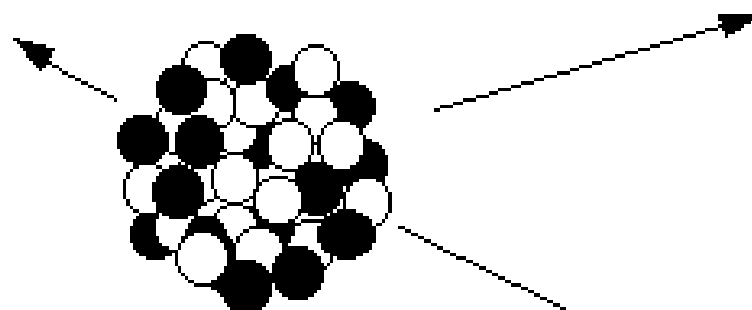
- Beta particles have a charge of **minus 1**. This means that beta particles are **the same as an electron**.
We can write them as β^- or e^- , because they're the same as an electron.
- This means that when a nucleus emits a β^- -particle: **the atomic mass is unchanged**, **the atomic number increases or decreases by 1**.
- They are **fast, and light**.
- Beta particles have a **medium penetrating power** - they are stopped by a sheet of **aluminium**.
- Example of radiopharmaceutical emits **phosphorus-32**.
- Beta particles ionise atoms that they pass, but not as strongly as alpha particles do.



Beta Particle Radiation



Daughter
Nucleus
Calcium-40



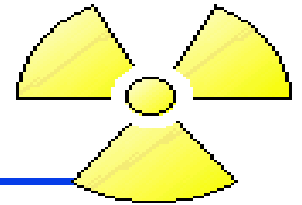
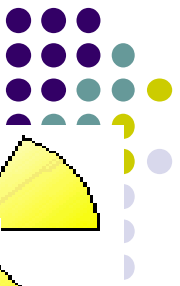
Parent Nucleus
Potassium-40

${}^0_0\bar{\nu}$

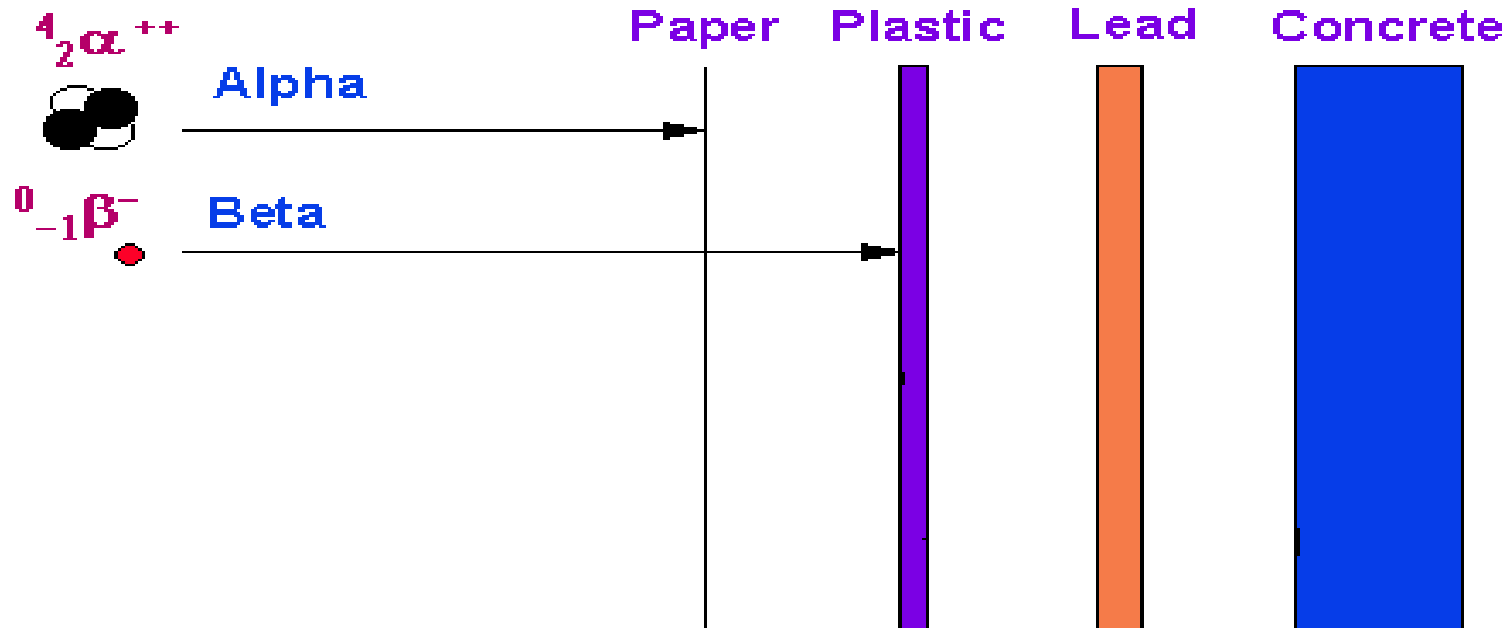
Antineutrino

${}^0_{-1}\beta^-$

Beta Particle



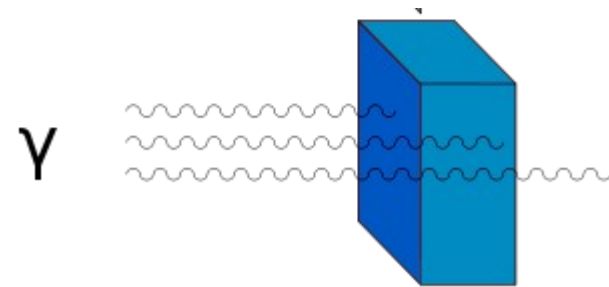
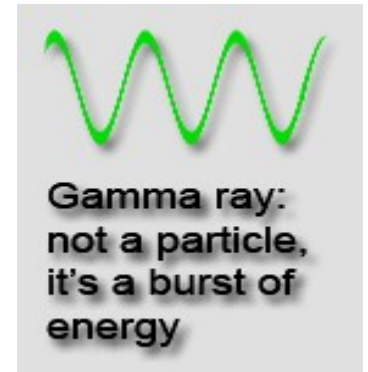
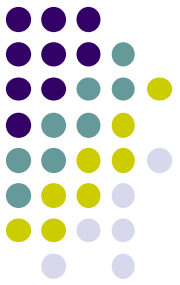
Penetrating Distances



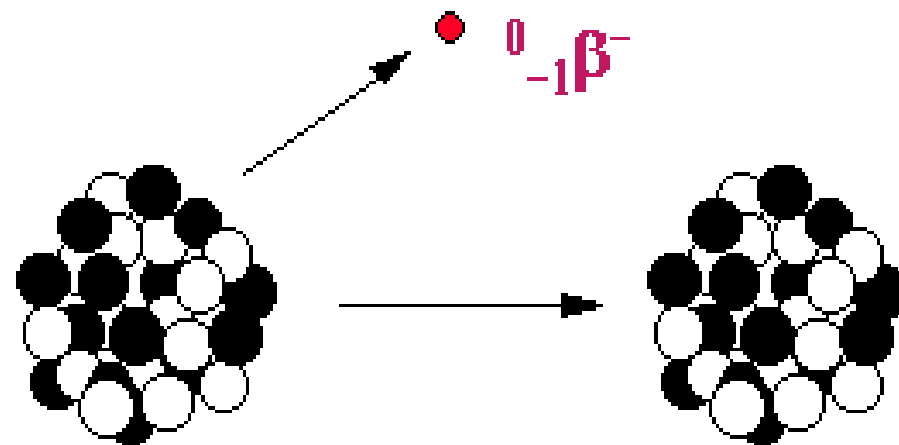
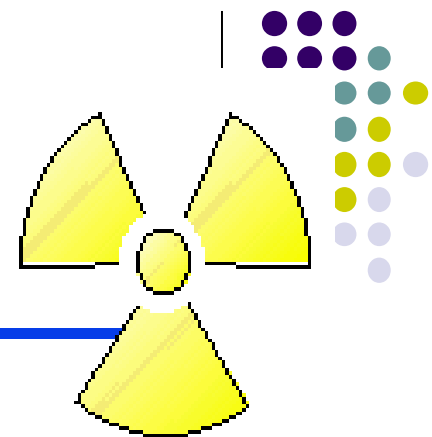
Beta particles are *much less massive and less charged* than alpha particles and *interact less intensely* with atoms in the materials they pass through, which gives them a longer range than alpha particles.

3- Gamma ray:

- Gamma rays are **waves, not particles**. This means that they have **no mass and no charge**.
- **in Gamma decay:**
 - **atomic number unchanged**
 - **atomic mass unchanged**.
- Gamma rays have **a high penetrating power**
 - it takes a thick sheet of metal such as **lead** to reduce them.
- Gamma rays do not directly ionise other atoms, although they may cause atoms to emit other particles which will then cause ionisation.
- We don't find pure gamma sources - gamma rays are emitted alongside alpha or beta particles.



Gamma-Ray Radiation

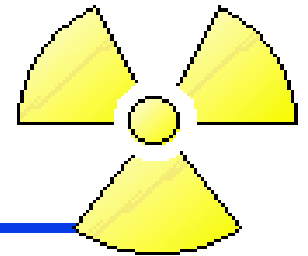
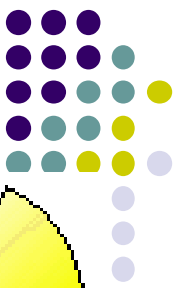


Parent Nucleus
Cobalt-60

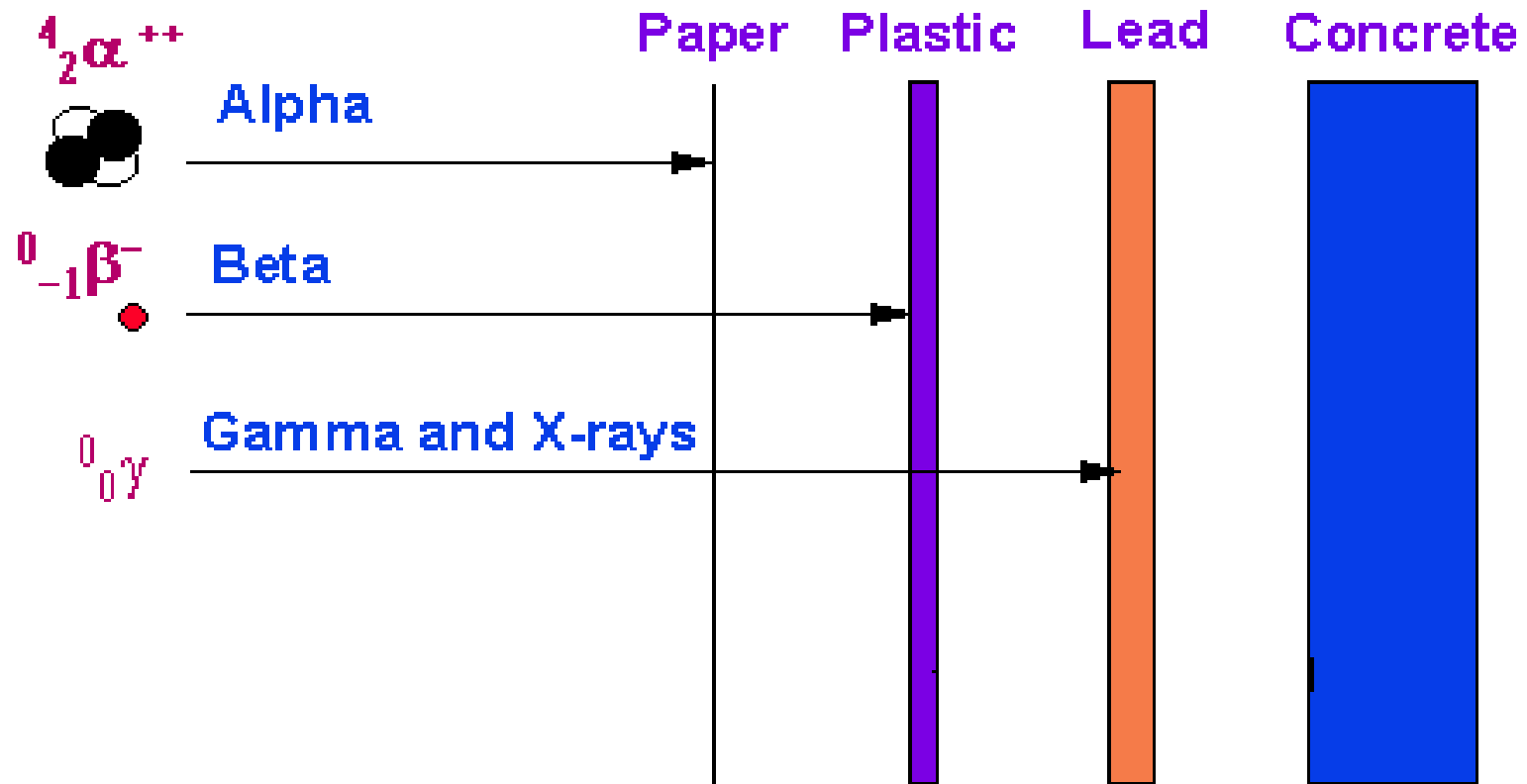
Daughter Nucleus
Ni-60

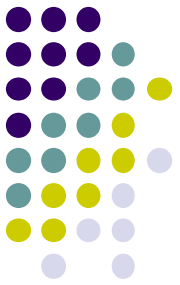


Gamma Rays



Penetrating Distances

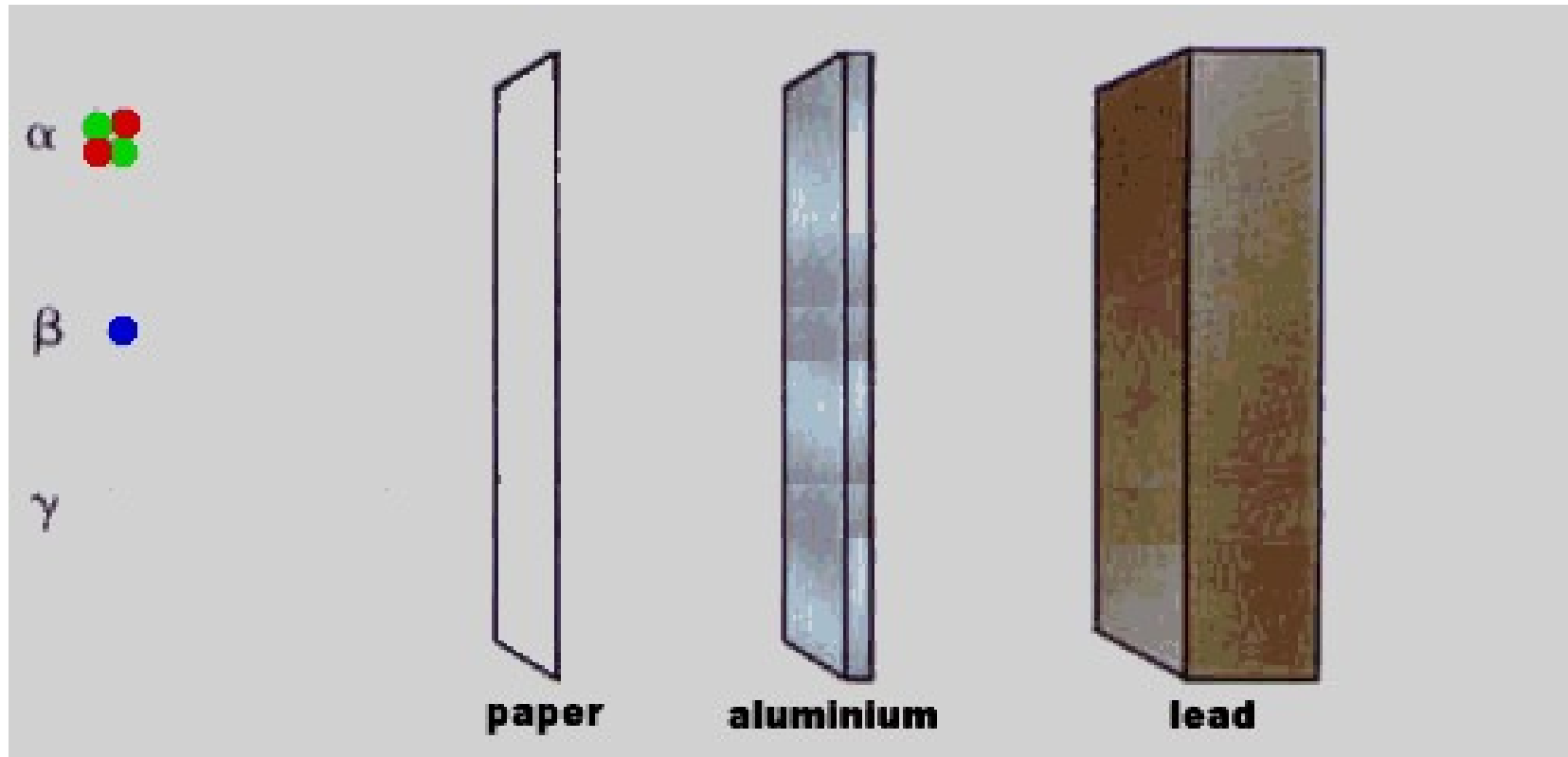
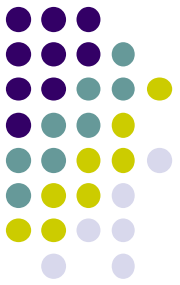


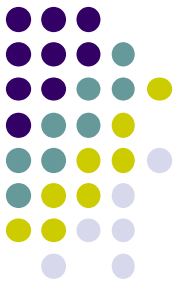


3- Gamma ray:

- Useful gamma sources include **Technetium-99m**, which is used as a "tracer" in medicine.
- This is a combined beta and gamma source, and is chosen because betas are less harmful to the patient than alphas (**less ionisation**) and because Technetium has a **short half-life** (just over 6 hours), so it decays away quickly and reduces the dose to the patient.

Alpha particles are easy to stop,
gamma rays are hard to stop.

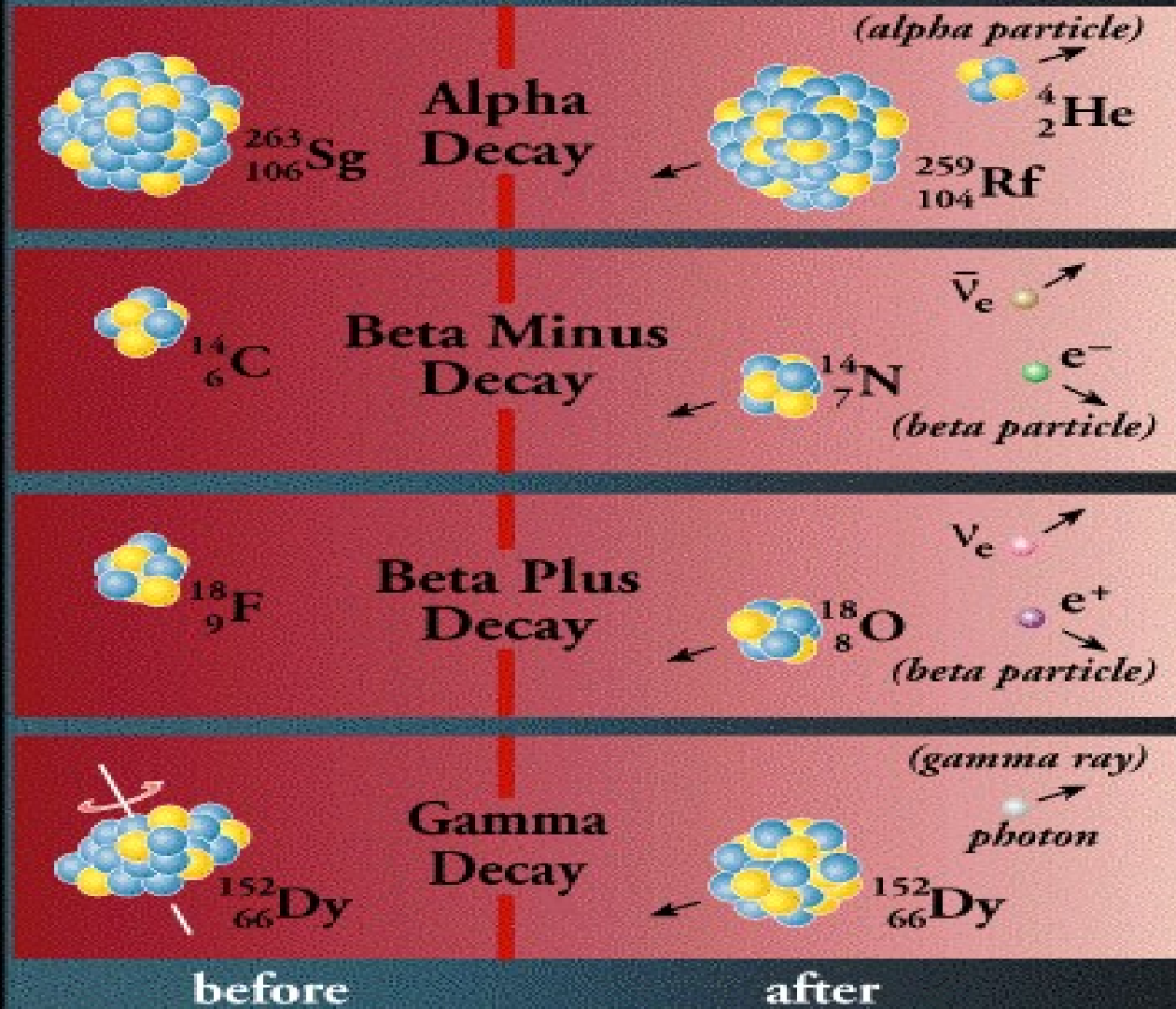
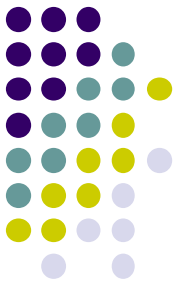


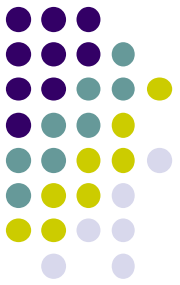


Mode of radioactive decay:

Type of Radiation	Alpha particle	Beta particle	Gamma ray
Symbol	${}^4_2\alpha$ or ${}^4_2\text{He}$	β^-	γ
Charge	+2	-1	0
Speed	slow	fast	Very fast
Ionising ability	high	medium	0
Penetrating power	low	medium	high
Stopped by:	paper	aluminium	lead

Radioactivity





Radiation measurement:

(*R*) *the roentgen for exposure:*

Is the amount of γ radiation that produces ionization of one electrostatic unit of either positive or negative charge per cubic centimeter of air at 0 °C and 760 mmHg.

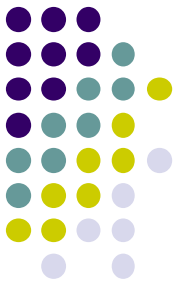
(*rad*) radiation absorbed dose is a more universal unit, it is a measure of the energy deposited in unit mass of any material by any type of radiation.

(*rem*) has been developed to account for the differences in effectiveness of different radiations in causing biological damage.

$$\text{Rem} = \text{rad} \times \text{RBE}$$

RBE is the relative biological effectiveness of the radiation.

Radiation measurement:



The basic unit for quantifying radioactivity (i.e. describes the rate at which the nuclei decay).

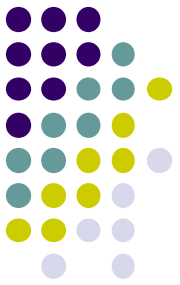
Curie (Ci):

- **Curie (Ci)**, named for the famed scientist Marie Curie
Curie = 3.7×10^{10} atoms disintegrate per second (dps)
Millicurie (mCi) = 3.7×10^7 dps
Microcurie (uCi) = 3.7×10^4 dps

Becquerel (Bq):

A unit of radioactivity. One becquerel is equal to 1 disintegration per second.

Properties of an Ideal Diagnostic Radioisotope:



- *Types of Emission:*

- Pure Gamma Emitter: (Alpha & Beta Particles are unimageable & Deliver High Radiation Dose.)

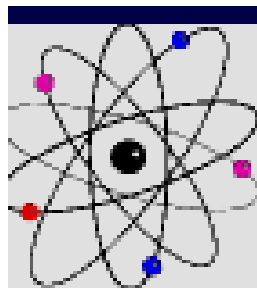
- *Energy of Gamma Rays:*

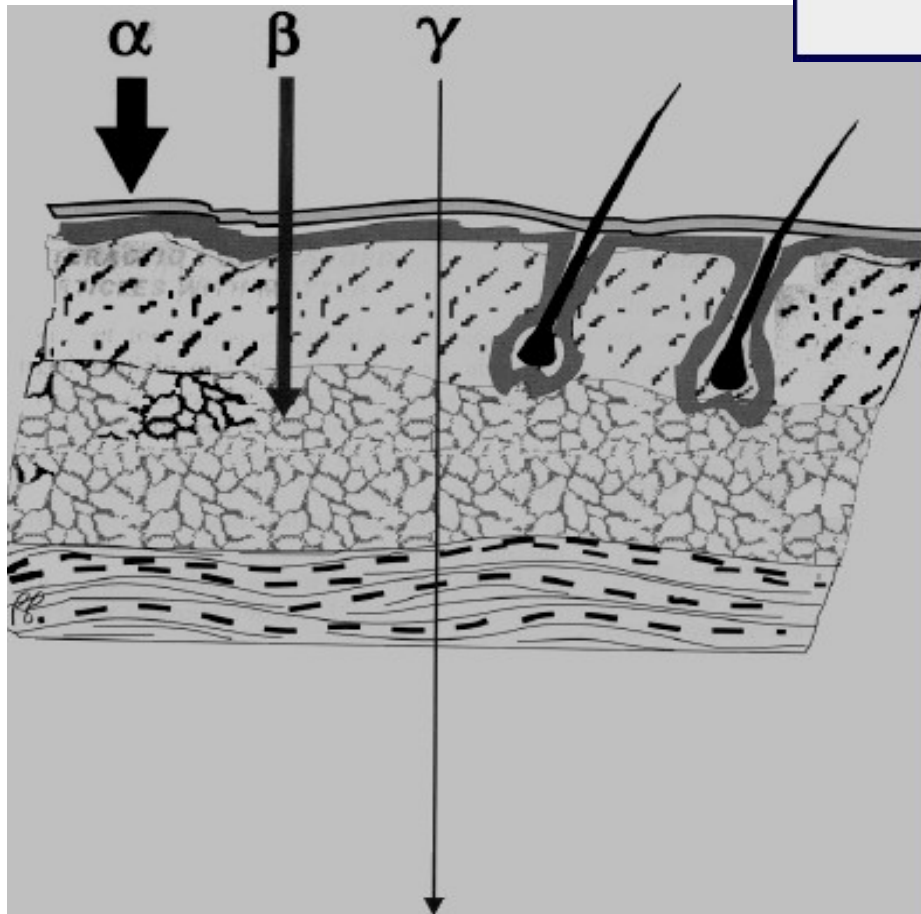
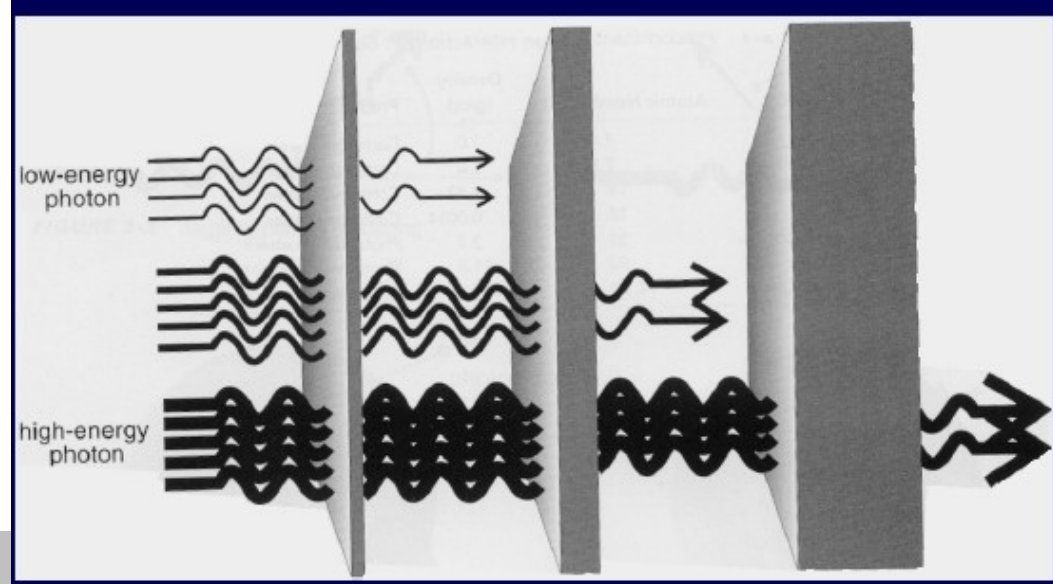
- Ideal: 100-250 keV e.g. ^{99m}Tc , ^{123}I , ^{111}In

- Suboptimal: <100 keV e.g. ^{201}Tl
>250 keV e.g. ^{67}Ga & ^{131}I

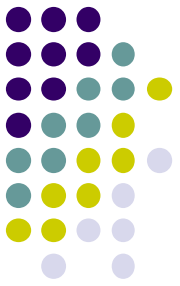
- *Photon Abundance:*

- Should be high to minimize imaging time





Properties of an Ideal Diagnostic Radioisotope:



- *Easy Availability:*

- Readily Available, Easily Produced & Inexpensive:

e.g. ^{11}C Vs $^{99\text{m}}\text{Tc}$

- *Target to Non target Ratio:*

- It should be high to:

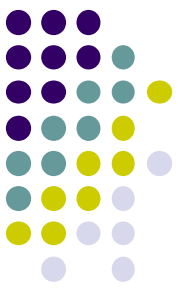
maximize the efficacy of diagnosis

minimize the radiation dose to the patient

- *Effective Half-life:*

- It should be short enough to minimize the radiation dose to patients and long enough to perform the procedure. Ideally 1.5 times the duration of the diagnostic procedure.

Properties of an Ideal Diagnostic Radioisotope:

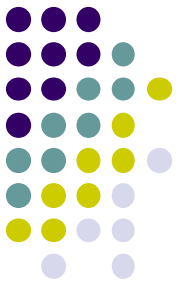


Example: For a Bone Scan which is a 4-h procedure, ^{99m}Tc -phosphate compounds with an effective half-life of 6 h are the ideal radiopharmaceuticals

- *Patient Safety:*
 - Should exhibit no toxicity to the patient.
- *Preparation and Quality Control:*
 - Should be simple with little manipulation.
 - No complicated equipment
 - No time consuming steps



Preparation of Radiopharmaceutical



1- Sterilization:

- Radiopharmaceutical preparations intended for parenteral administration are sterilized by a suitable method.
- Terminal sterilization by *autoclaving is recommended for heat stable products*
- For *heat labile products, the filtration* method is recommended.

2- Addition of antimicrobial preservatives:

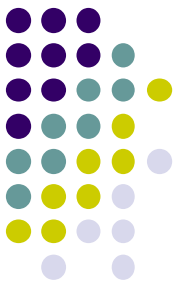
- Radiopharmaceutical injections are commonly supplied in multidose containers.

Preparation of Radiopharmaceutical :



- The requirement of the general monograph for parenteral preparations that such injections should contain a suitable antimicrobial preservative in a suitable concentration does not necessarily apply to radiopharmaceutical preparations.
- A reason for this exemption is that many common antimicrobial preservatives (*for example, benzyl alcohol*) are gradually decomposed by the effect of radiation in aqueous solutions.

3- Compounding:



- compounding can be **as simple as:**
 - adding a radioactive liquid to a commercially available reagent kit

- **as complex as:**

1- the creation of a multi-component reagent kit

N.B. Kit for radiopharmaceutical preparation

means a sterile and pyrogen-free reaction vial containing the nonradioactive chemicals [e.g., complexing agent (ligand), reducing agent, stabilizer, or dispersing agent] that are required to produce a specific radiopharmaceutical after reaction with a radioactive component.

2- the synthesis of a radiolabeled compound via a multi-step preparation process.

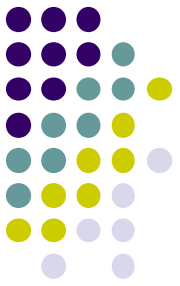
3- Compounding:



- The process of compounding radiopharmaceuticals must be under the supervision of recognized nuclear physician or a radiopharmacist.

- **STABILITY OF COMPOUNDED PREPARATIONS**

All extemporaneously compounded parenteral radiopharmaceutical preparations should be used no more than 24 hours post compounding process unless data are available to support longer storage.



Radiation shielding:

Adequate shielding must be used to protect laboratory personnel from ionizing radiation.





Pro-Tec II Syringe Shield



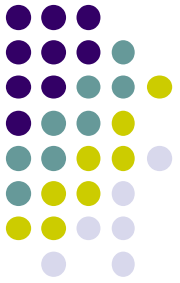
Guard Lock PET Syringe Shield



Pro-Tec V Syringe Shield

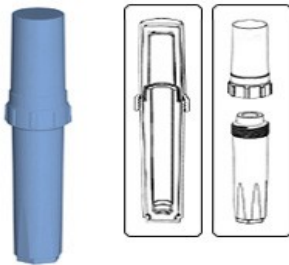


Color Coded Vial Shields





Vial Shield



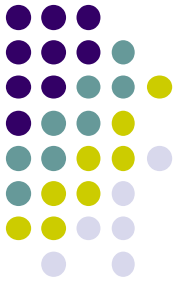
Unit Dose Pig

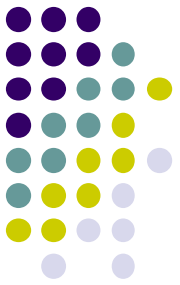


High Density Lead Glass Vial Shield



Sharps Container Shields





Radiation shielding:

- Alpha and beta radiations are readily shielded because of their limited range of penetration.
- The *alpha particles* are mono-energetic and have a range of a few centimetres in air.
- aluminium, glass, or transparent plastic materials, are used to shield sources of *beta radiation*.
- *Gamma radiation* is commonly shielded with lead and tungsten.

Radiopharmaceutical quality control:



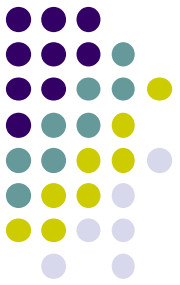
- ***Visual Inspection of Product***

- Visual inspection of the compounded radiopharmaceutical shall be conducted to ensure the absence of foreign matter and also to establish product identity by confirming that
 - (1) a liquid product is a solution, a colloid, or a suspension
 - (2) a solid product has defined properties that identify it.

- ***Assessment of Radioactivity***

- The amount of radioactivity in each compounded radiopharmaceutical should be verified and documented prior to dispensing, using a proper standardized **radionuclide (dose) calibrator.**

Radiopharmaceutical quality control:



- ***Radionuclidic Purity***

- Radionuclidic purity can be determined with the use of a suitable counting device
- The gamma-ray spectrum, should not be significantly different from that of a standardized solution of the radionuclide.

- ***Radiochemical purity***

- Radiochemical purity is assessed by a variety of analytical techniques such as:
 - liquid chromatography
 - paper chromatography
 - thin-layer chromatography
 - electrophoresis

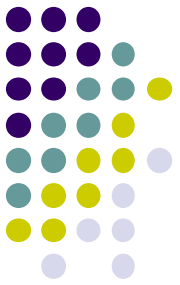
the distribution of radioactivity on the chromatogram is determined.

Radiopharmaceutical quality control:



- ***Verification of Macroaggregate Particle Size and Number***
- ***pH***
- ***Microbiological Control (sterility test) and Bacterial Endotoxin Testing***

Radiopharmaceutical quality control:

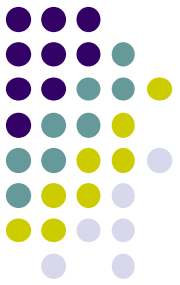


- **Labelling**

The label on the outer package should include:

- a statement that the product is radioactive *or* the international symbol for radioactivity
- the name of the radiopharmaceutical preparation;
- the preparation is for diagnostic or for therapeutic use;
- the route of administration;
- the total radioactivity present (for example, in MBq per ml of the solution)
- the expiry date
- the batch (lot) number
- for solutions, the total volume;
- any special storage requirements with respect to temperature and light;
- the name and concentration of any added microbial preservative

Application of radiopharmaceuticals:

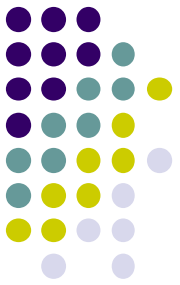


1- Treatment of disease: (therapeutic radiopharmaceuticals)

They are radiolabeled molecules designed to deliver therapeutic doses of ionizing radiation to specific diseased sites.

- *Chromic phosphate P32* for lung, ovarian, uterine, and prostate cancers
- *Sodium iodide I 131* for thyroid cancer
- *Samarium Sm 153* for cancerous bone tissue
- *Sodium phosphate P 32* for cancerous bone tissue and other types of cancers
- *Strontium chloride Sr 89* for cancerous bone tissue

Application of radiopharmaceuticals:



2- As an aid in the diagnosis of disease (diagnostic radiopharmaceuticals)

The radiopharmaceutical accumulated in an organ of interest emit gamma radiation which are used for imaging of the organs with the help of an external imaging device called gamma camera.

- Radiopharmaceuticals used in tracer techniques for measuring physiological parameters (e.g. ^{51}Cr -EDTA for measuring glomerular filtration rate).
- Radiopharmaceuticals for diagnostic imaging (e.g. $^{99\text{m}}\text{Tc}$ -methylene diphosphonate (MDP) used in bone scanning).